

House Price Prediction Using Machine Learning

XYlofy AI – Week 1 Internship Project

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Executive Summary

The objective of this project was to develop machine learning models capable of predicting house prices based on property characteristics and amenities. A complete data science workflow was followed, including data exploration, preprocessing, model development, evaluation, visualization, and business analysis. The project aimed to identify the most influential factors affecting house prices while building reliable predictive models.

Dataset Overview

The analysis was conducted using a housing dataset containing **545 property records** and **13 features** describing various aspects of residential properties. Data quality assessment confirmed that the dataset contained **no missing values** and **no duplicate records**, making it suitable for machine learning analysis.

Models Evaluated

Three regression models were developed and evaluated:

Model	R ² Score
Linear Regression	0.6529
Random Forest Regressor	0.6114
Tuned Random Forest Regressor	0.5854

Among all evaluated models, **Linear Regression achieved the best predictive performance**, explaining approximately **65% of the variation in house prices**.

Key Findings

- Property **area** was identified as the most influential factor affecting house prices.
- The number of **bathrooms** was the second most important predictor.
- Additional amenities such as **air conditioning**, **parking availability**, and the number of **stories** contributed significantly to property value.
- Unfurnished properties generally showed lower market prices compared to furnished alternatives.
- Despite testing more complex machine learning models, Linear Regression demonstrated the strongest performance on unseen data.

Business Recommendation

Real estate businesses should prioritize property size, bathroom count, parking facilities, and comfort-related amenities when estimating property values. These features have the greatest impact on pricing and can significantly influence buyer demand and investment decisions.

Conclusion

This project successfully demonstrated the application of machine learning techniques for house price prediction. Through data-driven analysis, key valuation factors were identified and evaluated using multiple regression models. The findings highlight the importance of property characteristics in determining market value and showcase how machine learning can support informed decision-making within the real estate industry.